QUIZ for Weeks 1-3 (EBU6230 2020-21)

The quizzes cover the introduction of image and video processing, image representation and image histogram. Please note that this online quiz will not be assessed and it will only available for a week.

分数: 9/15

1

Assume an image is quantized with 128 grey levels. How many bits are required to represent each grey level?
(2 分)

- 4
- 5
- 6
- 7

2

State TRUE or FALSE:
"A digital image can be represented by a matrix."
(1分)

- TRUE

- FALSE

3

State TRUE or FALSE:
"A color image CANNOT be represented by its color channels"
(1分)

- TRUE
- FALSE

4

For denoising an image, low-pass filters can be used and they remove high frequency components (noise). What effect will you see in the image after denoising?
(1分)

- The image will be clipped.
- The image will be shrinked.
- The image will lose edge details
- No change will be visible.

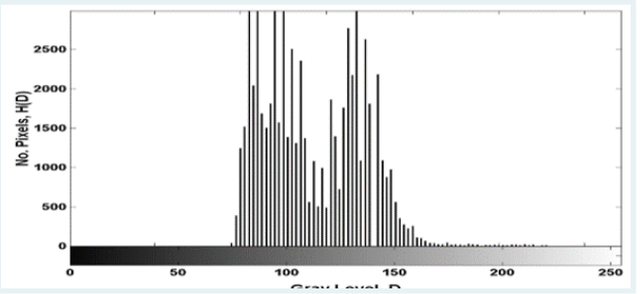
5

Name major factors that determine the digital image quality.
(2 分)

- Spatial resolution
- Intensity resolution
- Spatial sampling
- Number of quantization levels

All of the above

6



The figure on the right shows a histogram of an image, which can be used to reveal the image contrast. What does this histogram tell about the image? (1分)

- (●) This is a low-contrast image.
- ( ) This is a high-contrast image.
- ( ) This is a good-contrast image.
- ( ) Histograms do not reveal the image contras.

7

Why do we need to compress an image? (1分)

- ( ) To reduce image file size
- ( ) To reduce the transmission cost
- ( ) To reduce storage needed to store images
- (●) All of the above

8

State TRUE or FALSE:
"During quantization, the brightness sampled at each point in the continuous-tone image is replaced by an integer value. Generally, coarse quantization (i.e., quantization with fewer bits) often results in the appearance of abrupt discontinuities in the image (false contours). Dithering can be used to alleviate this problem."
(1分)

- (●) TRUE
- ( ) FALSE

9

State TRUE or FALSE:
"Denoising is used to remove noise and enhance the quality of an image."
(1分)

- (●) TRUE
- ( ) FALSE

10

What do you mean by High Dynamic Range (HDR) in an image? How can you create an HDR image?
(4 分)

HDR means very bright and very dark parts in a single image (many distinct intensity values). Cre

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由

High Dynamic Range (HDR) images

- High dynamic range means very bright and very dark parts in a single image (many distinct intensity values)
- Dynamic range in photographed scene may exceed number of available bits to represent pixels
- Solution:
  - Capture multiple images at different exposures
  - Combine them using image processing

High Dynamic Range (HDR) images

- Combine multiple images from same viewpoint
- Get the best of several different exposures